

This lecture provides an overview of tuples in Python, comparing and contrasting them with lists. Here's a breakdown of the key concepts:

- Introduction to Tuples:
 - Tuples are a data structure in Python, similar to lists, but with a key difference: immutability.
 - Tuples are represented using parentheses `()`.
 - Tuples can store various data types, including integers, floats, strings, complex numbers, and booleans.
- Similarities with Lists:
 - Tuples support indexing and slicing, just like lists and strings.
 - Elements can be accessed using their index (e.g., `T[0]`).
 - Slicing allows extracting a portion of the tuple (e.g., `T[2:8]`).
- Key Difference: Immutability:
 - The major difference between tuples and lists is that tuples are *immutable*. This means that once a tuple is created, its elements cannot be changed.
 - Attempting to modify an element in a tuple (e.g., `T[0] = 'new_value'`) will result in an error.
 - Lists, on the other hand, are *mutable*, allowing modification of their elements.
- When to Use Tuples:
 - Tuples are useful when you want to ensure that data remains constant and is not accidentally modified.
 - They provide a level of data integrity.
- Tuple Operations:
 - Iteration: You can iterate through a tuple using a `for` loop.
 - Conditional Logic: You can apply conditional statements to elements within a tuple during iteration (e.g., filtering for integers).
- Tuple Packing and Unpacking:
 - Packing: Creating a tuple by assigning multiple values at once (e.g., `T = (1, 2, 3)`).
 - Unpacking: Assigning the elements of a tuple to individual variables (e.g., `a, b, c = T`). This assigns `a = 1`, `b = 2`, and `c = 3`.
- Inbuilt Functions:
 - `count()`: Returns the number of times a specific element appears in the tuple.
 - `index()`: Returns the index of the *first* occurrence of a specific element.
- Nested Tuples and Lists within Tuples:
 - Tuples can contain other tuples (nested tuples).
 - Tuples can also contain lists as elements.
 - Elements within nested structures can be accessed using multiple indexing (e.g., `T[4][1]` to access the second element of a list located at index 4 of tuple `T`).
- Other Operations:
 - `len()`: Returns the number of elements in the tuple.
 - `for` loops, indexing, and slicing work similarly to lists.
- Key Takeaway:

- Tuples are immutable, while lists are mutable. Choose tuples when you need to ensure data integrity and prevent modification.
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